

Report R12 — C156 in the matrix: the sectors, drawn as block-diagonal form

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-08-04

Abstract

Every fragmentation statement in R2 and R9 is a number: a sector count, a growth base, a sub-leading exponent. Those numbers describe a property of a matrix — that the one-step transition matrix is block-diagonal in a basis nobody had drawn — and this report draws it. Two spy plots of the *same* matrix for rule 156 (IIVI) at obc0: once in the computational basis, once with the basis permuted so that each weak component is contiguous. The permuted picture is a staircase of 144 disjoint blocks at $N = 10$, the largest holding 20 of 1024 states, and **not one nonzero lies outside a block** — checked, at every N from 8 to 12, rather than asserted. Three quantitative readings come out of the same computation: the block count is exactly the Fibonacci number F_{N+2} , which is where R2's golden ratio comes from; the matrix thins from 1.81% to 0.24% dense between $N = 8$ and $N = 12$ while the *interior* of the blocks stays roughly two-thirds full, so the concentration factor grows from $41\times$ to $259\times$; and the computational-basis panel is emphatically not structureless — it is strongly banded and self-similar — it simply does not show *sector* structure. obc0 only; pbc is held.

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1 What is plotted, and why

Definition 1 (the matrix). For a rule, a length N and a boundary condition, let

$$M_{yx} = 1 \iff \text{the one-step map carries } |x\rangle \text{ to } |y\rangle \text{ with nonzero amplitude.}$$

M is the support of the circuit unitary U , equivalently of the doubly stochastic $|U|^2$; the two have the same pattern, which is all a spy plot uses. Amplitudes are deliberately discarded — the point of the figure is that the *support alone* already carries the fragmentation.

Definition 2 (the permutation). Let the weak components of the transition graph be $S_1, \dots, S_K$, ordered by size, largest first, with states ascending inside each. Let P be the permutation that lists the basis in that order. The second panel plots PMP^{-1} .

Why rule 156. It is unitary (IIVI: two identities, a Hadamard, an identity), so $|U|^2$ is doubly stochastic, there are no transient states, and the weak, strong and forward-closure partitions all coincide — R9’s assertion A2. The blocks are therefore unambiguous, and there is no question of whether we are drawing sectors or attractors: for this rule they are the same objects. It is also the rule R2 solves by room-packing, so its numbers are known independently of anything here.

Why it is worth a figure. “The dynamics block-diagonalises” is the central claim of the whole sector programme, and until now it has appeared only as counts and exponents. A permutation is not a theorem, but it makes the claim visible in one glance, and it makes the *failure* of the computational basis to reveal it equally visible.

2 The two panels

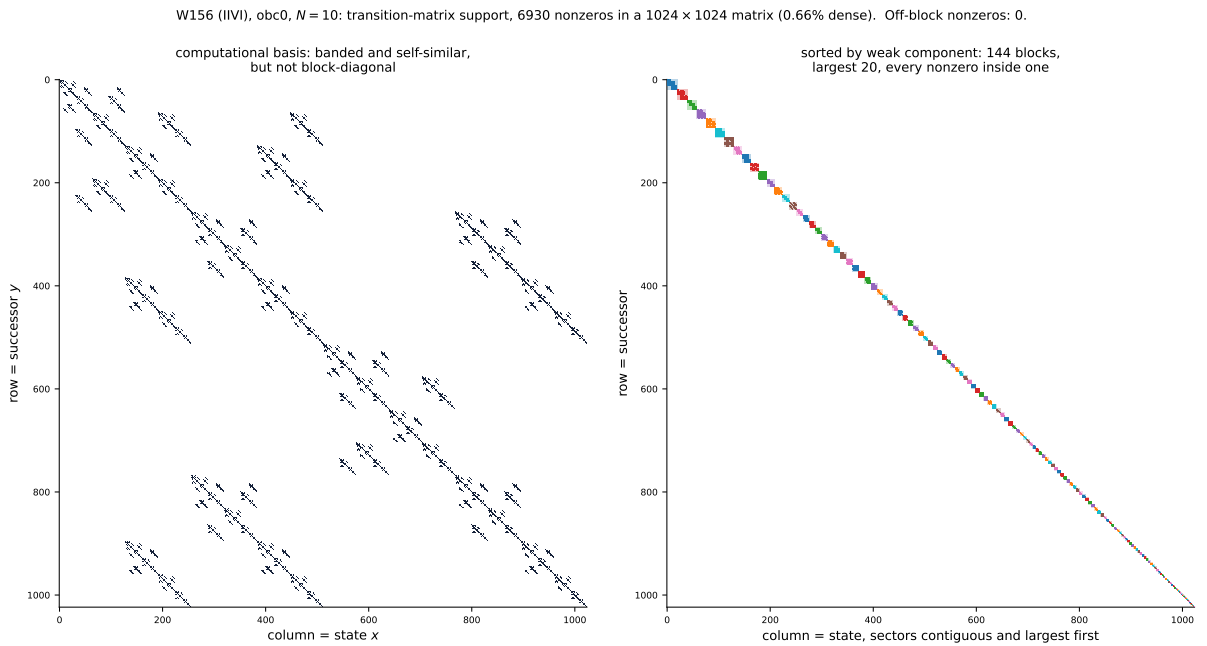


Figure 1: **S1, the same matrix in two bases** (rule 156, obc0, $N = 10$; 1024×1024 , 6930 nonzeros, 0.66% dense). **Left:** computational basis, i.e. states ordered by their integer value. **Right:** basis permuted so each weak component is contiguous, largest first; the shaded squares are the components and the dots are coloured by which component they belong to. The two panels contain *identical* data. The right panel’s caption number — off-block nonzeros = 0 — is computed, not assumed.

Three things to read off Figure 1.

- **The sorted panel is exactly block-diagonal.** Every one of the 6930 nonzeros lies inside one of the 144 shaded squares. Nothing is off-diagonal, at any N tested (§3).
- **The blocks are small and there are many of them.** The largest holds 20 of 1024 states, so the staircase is a thin diagonal thread rather than a few fat blocks. That is $D_{\max}/2^N \approx 0.02$ and falling — the quantitative form of “strong fragmentation”, and the reason R2’s $b = 4^{1/5}$ sits so far below the ergodic $b = 2$.
- **The left panel is *not* structureless.** It is strongly banded, with a self-similar block-of-blocks pattern at several scales. That is real and has an explanation — the integer order

groups states by their high bits, and the rule is local, so states that agree on most sites sit near one another — but the bands run *across* sectors. Ordering by integer value reveals locality, not conservation. The contrast between the panels is not structure-versus-noise; it is the wrong structure versus the right one.

2.1 A closer look

At $N = 10$ the largest block is 20 states wide, which is about four pixels in Figure 1, so the interior of a block cannot be seen there. Figure 2 takes the first 140 rows and columns.

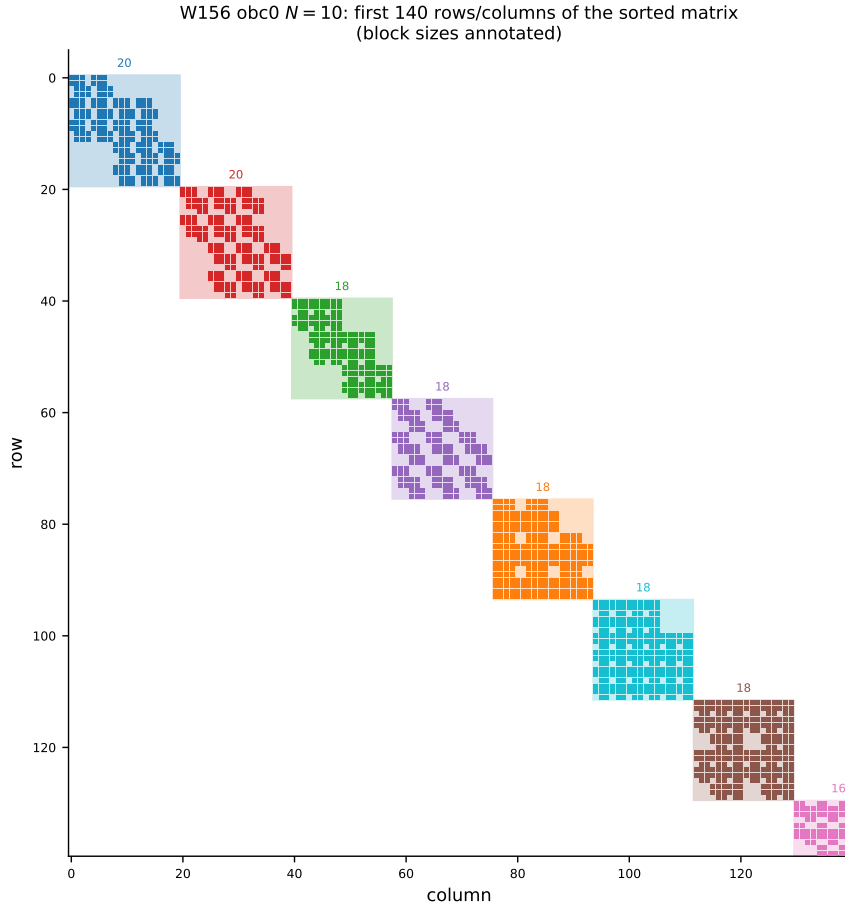


Figure 2: **S2, the top-left corner of the sorted matrix** (rule 156, obc0, $N = 10$), block sizes annotated. The largest sectors are 20, 20, 18, 18, 18, 18, 18, 16, $\dots$ states. Each block has its own internal sparsity pattern, itself visibly self-similar.

The blocks are *not* full: within a block the fill is about two-thirds (§3), and the interior pattern is itself structured rather than random. So a sector is not a featureless invariant subspace — it has its own sparse combinatorics, which is what the room-packing derivation of R2 is about. A smaller system makes the same point without a zoom:

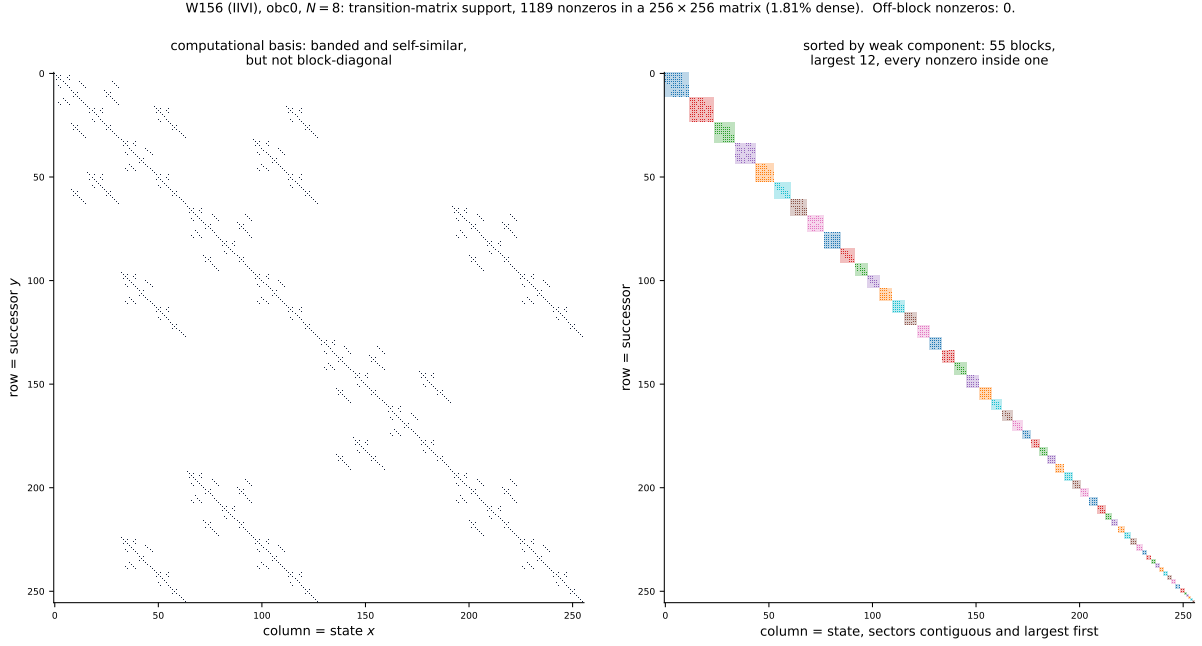


Figure 3: **S3, the same pair at $N = 8$** (256×256 , 55 blocks, largest 12). Small enough that the block structure is legible in the full matrix.

3 Numbers

N	dim	nnz	density	blocks	$D_{\max}$	off-block
8	256	1189	1.81%	55	12	0
9	512	2870	1.09%	89	16	0
10	1024	6930	0.66%	144	20	0
11	2048	16731	0.40%	233	27	0
12	4096	40391	0.24%	377	36	0

Off-block nonzeros are zero at every size. That column is the whole content of the sorted panel, and it is a computation (`spy.check_block_diagonal`), not a restatement of the definition: the blocks are found by weak connectivity on the transition graph, and the check then asks independently whether any matrix entry crosses between them. A nonzero there would mean the partition and the matrix disagree.

The block count is Fibonacci. For $N = 6 \dots 12$ the number of sectors is

$$21, 34, 55, 89, 144, 233, 377 = F_{N+2},$$

exactly, with $F_1 = F_2 = 1$. Hence the growth base $\varphi = 1.618034$ that R2 reports for this rule's sector count at obc0. (Under pbc the same rule gives $L_N + 1$ with L the Lucas numbers — a different sequence with the same base, which is why the boundary condition has to be stated with the constant.)

The nonzero count is a third constant, exactly. Counting the edges rather than the sectors gives, for $N = 6 \dots 14$,

$$204, 493, 1189, 2870, 6930, 16731, 40391, 97512, 235416,$$

which satisfies the integer recurrence

$$a_N = 2a_{N-1} + 2a_{N-3} + a_{N-4},$$

whose characteristic polynomial factors as

$$x^4 - 2x^3 - 2x - 1 = (x^2 - 2x - 1)(x^2 + 1).$$

The dominant root is the *silver ratio* $1 + \sqrt{2} = 2.414214$, and the $\pm i$ from the second factor is a pure period-4 modulation — which is exactly the $-1, -1, +1, +1$ pattern by which a_N misses $2a_{N-1} + a_{N-2}$. So this one rule carries three different algebraic constants at once, one for each question asked of the same matrix: φ for how many blocks there are, $4^{1/5}$ for how big the largest gets (R2), and $1 + \sqrt{2}$ for how many nonzeros the matrix has. None of the three is a fit; all are exact recurrences or derivations.

Thinning globally, not locally. As N grows the matrix becomes rapidly sparser in absolute terms, but the blocks do not hollow out:

N	density of M	fill <i>inside</i> the blocks	ratio
8	1.81%	74.1%	$41\times$
10	0.66%	68.0%	$103\times$
12	0.24%	62.3%	$259\times$

The middle column is the nonzero count divided by $\sum_k |S_k|^2$, the total area available inside the diagonal blocks. It drifts down slowly while the global density falls by a factor of 7.5, so essentially all of the sparsity of M is the block structure and almost none of it is sparsity within a sector. That is the precise sense in which the permutation is the right change of basis: it accounts for the emptiness.

4 What this does and does not show

- **It is support, not dynamics.** A spy plot cannot distinguish a sector that the dynamics explores ergodically from one it merely fails to leave. The block structure is a statement about which amplitudes can be nonzero, and nothing here measures how the weight moves inside a block.
- **It is one rule and one boundary condition.** Rule 156 was chosen because unitarity makes the partition unambiguous (A2). For a dissipative rule the same picture would need a choice — weak components block-*diagonal*, strong components only block-*triangular* — which is the distinction adjudicated in R-T14 and quantified in R9 §7.3. `pb` is held pending caveats and is not drawn.
- **The permutation is not canonical.** Blocks are ordered by size and states ascending within a block; any other order inside a sector gives an equally valid block-diagonal picture with a different interior pattern. What is canonical is the *partition*, and therefore the block sizes.
- **It does not prove the asymptotics.** The figure is a finite- N object. That the block count is F_{N+2} is checked to $N = 12$ here and derived in R2; the picture illustrates it and does not establish it.

5 Reproduce

```
python -m qca_fragmentation.viz.spy --rule 156 --N 10 --bc obc0
```

writes `fig_spy_156_N10_obc0`, `fig_spy_156_N8_obc0`, `fig_spy_zoom_156_N10_obc0` (each as `.pdf` and `.png`) and `reports/tex/tab_r12_spy_obc0.tex`. The module is `code/qca_fragmentation/viz/spy.py`; its tests are in `code/qca_fragmentation/tests/test_spy.py`, and they pin the zero off-block count, the Fibonacci block count, the partition agreeing with the Tier-1e `weak.components` pass, and the permutation being a genuine permutation.